

Low Level Design (LLD)

Energy Document Intelligence - RAG System

1. Introduction

This document provides the Low Level Design for the Energy Document Intelligence system based on Retrieval-Augmented Generation (RAG). The system allows users to upload energy-related documents (PDFs, manuals, reports) and perform context-aware question answering grounded strictly on the uploaded content.

2. Detailed Architecture Components

The system consists of the following internal modules:

- Document Upload Service – Handles PDF upload and validation.
- Text Extraction Module – Extracts raw text from PDFs.
- Text Chunking Module – Splits extracted text into manageable chunks.
- Embedding Service – Converts text chunks into vector embeddings using an LLM embedding API.
- Vector Database – Stores embeddings and metadata for similarity search.
- Query Processing Module – Converts user question into embedding.
- Retriever – Performs similarity search on Vector DB.
- LLM Answer Generator – Generates final answer using retrieved context.
- Response Formatter – Formats and returns final response to user.

3. End-to-End Process Flow

Step 1: User uploads an energy document (PDF).
Step 2: System extracts text from the document.
Step 3: Text is divided into smaller chunks.
Step 4: Each chunk is converted into vector embeddings.
Step 5: Embeddings are stored in Vector Database.

Step 6: User submits a question.
Step 7: Question is converted into embedding.
Step 8: Vector DB performs similarity search.
Step 9: Top relevant chunks are retrieved.
Step 10: Retrieved context + question are sent to LLM.
Step 11: LLM generates context-grounded answer.
Step 12: Final response is returned to user.

4. System Flow Diagram

